

A PAIRED FIXED-FILLER DIAGNOSTIC FOR A LENGTH-ASSOCIATED SCORE–OUTCOME DIFFERENCE

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ABSTRACT

A score recorded when a masked-diffusion decoder commits a token and the correctness of its final answer are measurements at different levels. We study their cross-level mean difference and ask a narrow diagnostic question: does one source-described fixed-filler intervention yield a change comparable in magnitude to a separate natural prompt-length analysis? The source manuscript reports a natural-strata difference of $+0.2227$ and, on a different 400-item pool, a paired $+37$ -token, fixed-NFE=64 filler contrast of -0.0075 with interval $[-0.052, +0.039]$. The latter point estimate is descriptively smaller in magnitude. The samples and estimands differ, however, and no joint comparison was performed. The source also reports increases of $+0.0213$ and $+0.0452$ in quantities labelled Brier and NLL. Because the current package does not establish that the commit score is an answer-correctness probability, we treat these as numerical score–outcome diagnostics. Every empirical number in this paper is source-reported and cannot be independently verified from the current package; no experiment was rerun for this revision.

1 INTRODUCTION

Prompt length can covary with task content, formatting, and difficulty. A natural length association therefore does not identify what a length-only intervention would do. This distinction is especially important for masked-diffusion decoding, where prompt construction, the unmasking trajectory, and a fixed inference budget can interact. We examine one source-described GSM8K (Cobbe et al., 2021) / LLaDA-8B-Instruct (Nie et al., 2025) record that places two analyses side by side: an independent natural-strata analysis and a paired fixed-filler analysis.

The central quantity is a *cross-level mean score–outcome difference*: the mean of a decoder’s recorded token-commit score minus mean final-answer correctness. The source artifacts call this quantity SCE; we retain that string only as an internal artifact alias. We do not call the score confidence or an answer probability because the package contains neither the token-to-answer aggregation nor construct-validation evidence. The quantity remains arithmetically well defined as a directional summary, but its zero point is not a calibration certificate (Ao et al., 2023; Seligmann et al., 2023; Cruz et al., 2024).

The study’s actual role is diagnostic. It asks whether one fixed filler operation can supply an operational account of the reported length-associated difference by producing a comparably large change on a recorded paired pool. The design does not identify the cause of the natural association, and the available summaries do not support an inferential comparison between the two analyses. The paper contributes:

1. a finite-pool estimand conditional on the recorded single-decode realization;
2. a compact, source-reported comparison of the natural-strata and paired-filler records with their different samples and targets made explicit; and
3. an evidence boundary that treats score-based losses and MATH500 judge variants as secondary numerical diagnostics rather than validated answer-level calibration or cross-bed confirmation.

Measurement status. Throughout, SR/CV means *source-reported and not independently verifiable from the current package*. Every empirical numerical value, interval, count, threshold, and dose in the manuscript is SR/CV. This global label also applies when space prevents repetition in individual cells. Evidence IDs A32-E1–A32-E4 are opaque references to missing experimental records, not claims that those records were rerun or recovered.

2 QUESTION, HIERARCHY, AND EVIDENCE BOUNDARY

The question is deliberately narrower than whether prompt length has a causal effect: under the source-described recorded prompts and decode realization, what paired change in the cross-level mean score–outcome difference accompanies the fixed filler? The natural-strata result supplies motivation and descriptive context rather than a second endpoint or a common-estimand comparator.

Table 1: Canonical evidence hierarchy. Every numerical entry is SR/CV.

Role	Analysis	Evidence	Interpretation boundary
Context	Natural prompt-length strata	A32-E1	Independent sample and observational target
Central diagnostic	Fixed-filler paired contrast	A32-E2	Finite pool; one recorded decode per item–arm
Secondary	Source-labelled Brier and NLL changes	A32-E3	Numerical score–outcome diagnostics only
Ancillary	MATH500 judge conventions	A32-E4	Descriptive scenarios; no cross-bed inference

What is missing. The authoring package does not contain the literal filler string, insertion location, tokenizer-level dose check, immutable 400-item manifest or selection rule, overlap map with the natural-strata sample, exact prompt serialization, arm execution order, token-to-answer score construction, or gate implementation. It also lacks per-item records, decode traces, preregistration, analysis code, model/tokenizer revision, environment lock, and a claim-to-file crosswalk. Consequently, the treatment, sampling frame, score construct, and source-reported intervals cannot be independently reconstructed.

3 FINITE-POOL RECORDED-REALIZATION ESTIMAND

Let $\mathcal{P}$ denote the source-described finite paired pool and $a \in \{S, L\}$ the Short and Long arms. Let ω denote the complete recorded single-decode realization, including every stochastic choice used to produce the stored arm outputs. For item i , let $s_{ia\omega}$ be the source-recorded commit score and $y_{ia\omega} \in \{0, 1\}$ final-answer correctness under the source’s matcher. Define the arm summary

$$D_{\mathcal{P},\omega}(a) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} (s_{ia\omega} - y_{ia\omega})$$

and the paired filler contrast

$$\Delta_{\mathcal{P},\omega} = D_{\mathcal{P},\omega}(L) - D_{\mathcal{P},\omega}(S) = \frac{1}{|\mathcal{P}|} \sum_{i \in \mathcal{P}} [(s_{iL\omega} - y_{iL\omega}) - (s_{iS\omega} - y_{iS\omega})].$$

This is a descriptive contrast for the named finite pool and stored decode realization. It is not an expectation over a problem population, decoder randomness, filler strings, or prompt constructions. A decoder-expectation estimand would require deterministic replay evidence or multiple paired decode realizations; neither is available.

The independent natural-strata analysis instead compares high- and low-length groups formed from a different source-reported record. Natural prompt content and difficulty co-vary with those groups. Because the package lacks item identities and an overlap map, no common-sample contrast, covariance-aware difference, equivalence margin, or joint test can be formed.

Uncertainty record. The source manuscript describes problem-level cluster bootstrap intervals. The resampling code, input rows, seeds, and replicates are absent, so we transcribe the reported brackets but cannot verify their coverage, family membership, or arithmetic. Even if those brackets are accepted as reported, they do not include decoder-run variation beyond ω .

4 SOURCE-REPORTED RESULTS

4.1 INDEPENDENT NATURAL-STRATA CONTEXT

The source reports that the high-minus-low natural-length contrast is $+0.2227$ in the cross-level mean difference. Its component accounting is an identity: the reported score change is -0.0314 and the reported accuracy change is -0.2541 , so $-0.0314 - (-0.2541) = +0.2227$. This verifies transcription arithmetic only, not the underlying measurements.

Table 2: SR/CV independent natural-strata record (A32-E1). The groups are observational and are not the paired filler pool.

Quantity, high minus low	Point	Reported bracket
Mean recorded commit score	-0.0314	$[-0.0396, -0.0236]$
Mean answer correctness	-0.2541	$[-0.3107, -0.1987]$
Cross-level mean score–outcome difference	$+0.2227$	$[+0.169, +0.277]$

4.2 PAIRED FIXED-FILLER DIAGNOSTIC

The source describes a 400-item paired comparison in which Long adds a fixed 37-token filler to Short under reported NFE=64. Table 3 transcribes the available summaries. The literal filler, insertion position, tokenizer verification, sample construction, and per-item pairing records are missing, so these labels describe the source record rather than an independently instantiated intervention.

Table 3: SR/CV Short/Long record (A32-E2–A32-E3). Brackets are transcribed from the source and cannot be recomputed from this package.

Quantity	Short	Long	Long minus Short
Cross-level mean score–outcome difference	-0.0724	-0.0799	$-0.0075 [-0.052, +0.039]$
Mean recorded commit score	0.630	0.543	$-0.0875 [-0.098, -0.076]$
Mean answer correctness	0.703	0.623	$-0.0800 [-0.125, -0.033]$
Brier-labelled numeric diagnostic	0.1922	0.2135	$+0.0213 [0.011, 0.031]$
NLL-labelled numeric diagnostic	0.5723	0.6175	$+0.0452 [0.024, 0.067]$

The source-reported filler point estimate, -0.0075 , is descriptively smaller in absolute magnitude than the independently reported $+0.2227$ natural-strata estimate. The comparison stops there. The two values concern different item sets and estimands, and no joint uncertainty calculation or pre-specified comparison rule is available. No inferential conclusion about their difference follows.

Operationally, the paired analysis diagnoses whether this one recorded fixed-filler transformation produces a comparably large finite-pool change. It does not determine whether filler, semantic content, task difficulty, formatting, position, or another factor explains the natural length association. A factorial intervention and an aligned common-sample analysis would be needed to separate those possibilities.

Score-based loss diagnostics. The source labels two transformations Brier and NLL and reports positive Long-minus- Short changes in Table 3. The score-extraction rule, answer-level forecast event, NLL clipping, and implementation are absent. We therefore treat these values only as numerical diagnostics of the recorded score–outcome pairs. They do not establish answer-level calibration or probabilistic quality.

Omitted random-dose analysis. The source manuscript also describes a Random arm and a dose regression. The row-level dose data, regression equation, design matrix, and code are absent, and the prior prose gave mutually ambiguous descriptions of its response and intercept. We omit the regression, dose bins, and associated inferential language rather than reconstructing them from summary numbers.

5 ANCILLARY MATH500 SCENARIO

MATH500 is retained only as an ancillary descriptive judge-convention scenario (A32-E4). The source reports an answer-aware convention value of $+0.2382$ with bracket $[+0.1639, +0.3125]$ and a lenient convention with a purposive-anchor adjustment of $+0.1606$ with band $[-0.059, +0.380]$. All four numbers are SR/CV. The second band is an anchor-dependent scenario range, not a confidence interval: its 30 anchors were purposively selected, and neither a sampling frame nor an identified sampling standard error is available.

The judge outputs, human labels, exact oracle construction, and correction code are absent. We therefore use neither convention for inference, confirmation, replication, or transport. The scenario records only that the source-reported reading changes with the answer-matching convention.

6 CLOSEST NEIGHBORING WORK AND BOUNDARY

Prior work establishes that prompt choices can change calibration-related behavior, but the verified neighboring studies do not test this exact operation. Xia et al. (Xia et al., 2025) study calibration across four prompt styles; their analysis does not isolate prompt length, content-free filler, masked-diffusion commit scores, or a same-item $+37$ -token intervention. Zhao et al. (Zhao et al., 2024) study six prompting strategies and propose a confidence-calibration prompting method; the prompting strategy and elicited confidence change, rather than a semantics-preserving length-only filler under fixed diffusion compute. Kamoda et al. (Kamoda et al., 2023) show that prompt variations affect factual-probing outputs and evaluate prompt augmentation as a calibration intervention; they use prompt variants and ensembling for factual probing, not the present fixed-NFE masked-diffusion filler estimand.

For measurement context, signed mean score-label summaries and their cancellation limits predate this study (Ao et al., 2023; Seligmann et al., 2023; Cruz et al., 2024). The present contribution boundary is therefore an operation-specific diagnostic record, not a new metric, a general theory of prompt length, or a verified answer-calibration result. A broader prompt-length-specific and masked-diffusion uncertainty search remains incomplete.

7 LIMITATIONS, AVAILABILITY, AND DISCLOSURE

Scientific limits. The record covers one reported model-task setting, one fixed filler description, one dose, one inference budget, and one stored decode per item-arm. The exact sample and intervention are missing. The observational and paired analyses use different targets; the score construct is not verified as an answer probability; the source intervals do not include decoder-run uncertainty; and the ancillary MATH scenario depends on missing judge artifacts and purposive anchors. These limits preclude causal, decoder-average, cross-model, cross-task, or transport conclusions.

What the current package contains. The authoring directory inspected for this revision contains `main.tex`, `refs.bib`, the ICLR style files, compiled manuscript and auxiliary files, and two pre-rendered figure assets. This revision includes no figure in the manuscript. The directory does not contain experimental records E1–E4, literal prompts, item manifests, preregistration, protocol or analysis code, judge or anchor files, model/tokenizer revisions, decode configuration, an environment lock, or an evidence-ID crosswalk. No empirical value was rerun or independently recomputed.

AI assistance disclosure. AI tools assisted with language editing, $\text{\LaTeX}$ restructuring, and isolated internal pre-submission review for this revision. The authors remain responsible for every claim and

evidence boundary. AI assistance did not supply experimental records, generate new measurements, verify the source-reported values, or create scientific figures for this revision.

8 CONCLUSION

The source manuscript reports a finite-pool, recorded-single-decode filler contrast of -0.0075 with bracket $[-0.052, +0.039]$ in the cross-level mean score–outcome difference. Its absolute point magnitude is descriptively smaller than the independently reported natural-strata value of $+0.2227$. Because the samples and estimands differ and no joint test exists, this juxtaposition is descriptive only. The paired record serves as a diagnostic of whether one fixed filler operation can account operationally for the length-associated gap; it does not identify or adjudicate the causal origin of that association. The Brier- and NLL-labelled values remain numeric score–outcome diagnostics, and MATH500 remains an ancillary judge-convention scenario. Missing records, construction details, score semantics, and decoder-replay evidence prevent stronger conclusions.

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